

City of Wander: Visualizing Scientific Literature for Knowledge Exploration Using Visual Metaphors

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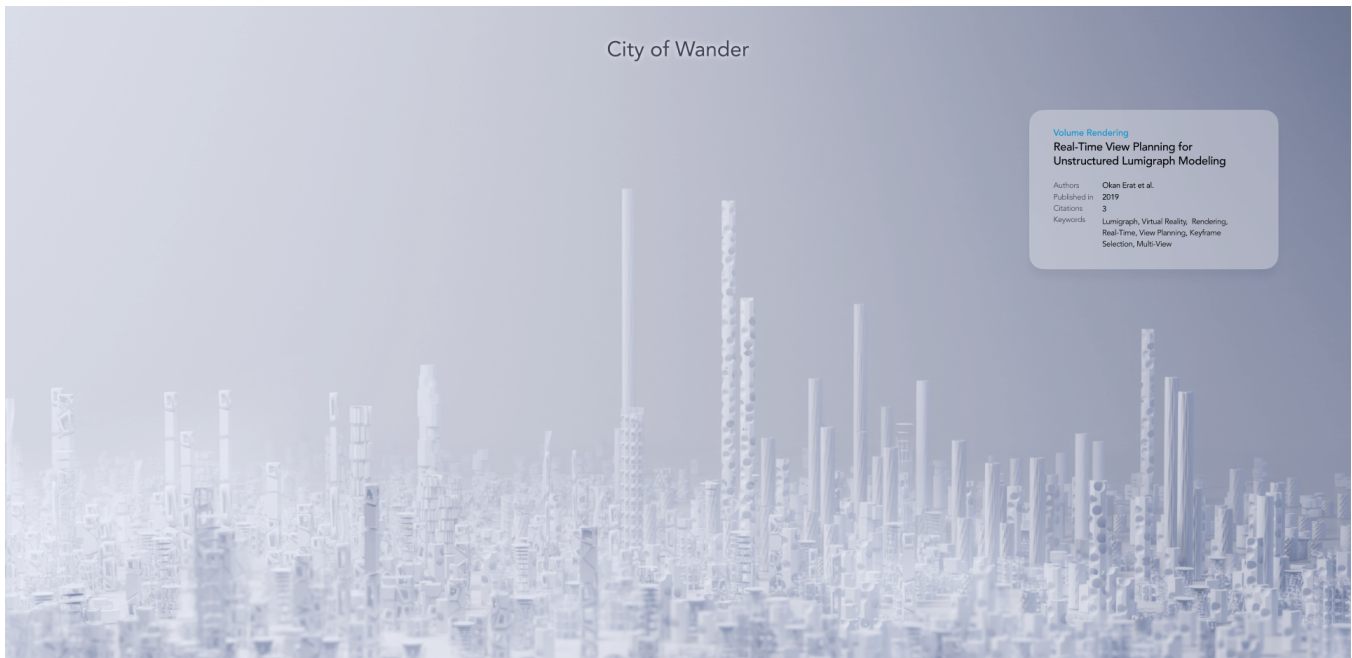


Figure 1: In City of Wander, each building represents a scientific paper from IEEE TVCG (1995–2023). Themes define the architectural forms, citation counts determine the building heights, and publication years influence the curvature. Roads symbolize citation relationships, weaving a network of intellectual connections. Navigate this dynamic urban landscape from a bird’s-eye view for thematic insights or from a ground-level perspective for immersive exploration.

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3720280>

Abstract

Scientific literature is essential for knowledge acquisition but can be challenging for non-expert users due to its technical complexity. Traditional visualization methods like word clouds, topic analysis, and network graphs provide insights but lack accessibility for non-experts. This paper proposes a novel approach to scientific literature visualization, using an architectural-city metaphor to create an immersive “Literature City” for exploring citation networks. Our system, City of Wander, combines data from IEEE Transactions

on Visualization and Computer Graphics (TVCG) (1996–2023), mapping it to parametric architectural forms for an intuitive, interactive experience. Two perspectives support diverse needs: the Bird’s-Eye Perspective offers a macroscopic view, while the Ground-Level Perspective enables immersive exploration of buildings and detailed literature. We evaluated City of Wander through a case study in the Immersive Reality field and informal exhibition testing, demonstrating its potential to engage and simplify complex information for non-experts.

CCS Concepts

• **Human-centered computing** → **Visualization design and evaluation methods.**

Keywords

Knowledge exploration, Scientific literature, Visual metaphor, Visualization design

ACM Reference Format:

Guanhong Liu, Yutong Jiang, Xiaoke Yan, Nan Cao, and Yang Shi. 2025. City of Wander: Visualizing Scientific Literature for Knowledge Exploration Using Visual Metaphors. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3706599.3720280>

1 Introduction

Scientific literature is a valuable knowledge resource that helps readers quickly grasp key concepts in specific fields. However, due to its technicality and complexity, it can be challenging to understand. Traditional visualization methods, such as word clouds [10], topic content analysis [4], and network graphs [25], visualize the evolution of themes in the literature and the relationships between publications. These methods assist users in identifying research trends, analyzing problems, and predicting future directions. However, they are often not user-friendly for non-experts who lack domain knowledge and skills, which limits their ability to engage with scientific literature.

In recent years, researchers have focused on designing more intuitive and engaging visualization tools to help non-experts understand complex data [2, 21]. Current scientific literature visualizations are mostly based on citation networks, such as Citeology [20] and CiteSpace II [24], which are primarily designed for expert users. These visualizations often take the form of abstract charts and complex data structures, lacking in attractiveness and intuitiveness. To democratize the exploration of scientific literature, it is necessary to introduce visual metaphors [3, 23] and immersive interactions [6, 13, 17, 19], using figurative and concrete representations to make complex scientific information more engaging and participatory, thereby sparking users’ curiosity and desire to explore.

Thus, we propose an innovative approach to scientific literature visualization, supporting knowledge exploration within scientific literature through visual metaphors. We presented City of Wander, an immersive visualization system. By combining the citation network of scientific literature with an architectural-city metaphor, we created a virtual “Literature City” that allows users to explore the relationships between documents and acquire new knowledge.

The data used in this study comes from articles published in IEEE Transactions on Visualization and Computer Graphics (TVCG) from 1996 to 2023, where literature data is mapped to architectural forms through parametric design. To evaluate City of Wander, we conducted a case study on the Immersive Reality field, demonstrating its capability to reveal temporal and thematic trends in research. Also, City of Wander was featured in an exhibition, where non-expert participants engaged with the system in a VR setting. Our work aims to provide insights for future research in scientific literature visualization, particularly in how to make complex scientific information more attractive and engaging through visualization and concrete representations, thereby stimulating users’ curiosity and desire to explore.

Our contributions are two-folds as follows:

- We propose the use of architectural-city metaphors to visualize scientific literature, offering an intuitive and engaging alternative to traditional abstract visualizations. This approach enhances knowledge acquisition for non-experts by fostering curiosity and stimulating exploration interest.
- We present City of Wander, a VR-based prototype that implements this metaphor. By mapping citation networks to a virtual “Literature City”, it enables immersive exploration of research relationships, providing a novel, interactive way to navigate scientific knowledge.

2 Related Work

Our work builds on prior research on visualizing citation networks and visual metaphors in visualization design.

2.1 Visualizing Citation Networks

Citation networks are a valuable tool for analyzing scientific literature, revealing the influence of papers [27], aiding the search for related works [12], and supporting thematic exploration [16]. Citation data is commonly visualized using node-link diagrams, which provide an overview of paper relationships, allowing users to quickly grasp key trends before diving into more detailed analysis. These diagrams also facilitate exploration by enabling users to start from points of interest and follow related connections. For instance, Citeology [20] uses node-link diagrams combined with stacked bar charts to represent the citation relationships between publications. CiteSpace II [24] visualizes the interactions between research frontiers and their knowledge foundations through citation networks. To improve the layout and reduce overlaps, PaperVis [7] modifies radial space-filling and target-view techniques, placing key publications at the center and using a radial node-link layout. Some studies, such as CiteVis [22], focus on task-specific layouts, replacing node-link diagrams with attribute-based designs to clearly display citation and reference relationships. Our work builds upon these visualization techniques by combining citation network layouts with an innovative metaphorical approach to enhance engagement and accessibility for non-experts.

2.2 Visual Metaphors in Visualization Design

Visual metaphors offer an intuitive way to represent complex data by relating unfamiliar concepts to familiar objects, stimulating curiosity, often incorporating interactive and aesthetic elements.

For example, Latvala et al. [18] used an aquarium metaphor to represent network data, with nodes depicted as fish within a 3D space. Similarly, SkyGlyphs [1] uses animated balloons to visualize data points, drawing attention with their movement and vibrant colors. Metaphors can also evoke emotional responses or deeper reflections by tying data to cultural or symbolic imagery. For instance, Pedro et al. [9] used tree rings to visualize U.S. immigration data, linking population growth to tree growth patterns. Tied in Knots [14] employed metaphor to convey empathy for sexual assault victims, while An Ecosystem of Corporate Politicians [8] used insect metaphors to satirically depict corporate politics. Architectural and urban metaphors are frequently used to express abstract or non-spatial data. Vincur et al. [5] applied urban metaphors to represent software systems, using architectural layouts to depict metrics and couplings. Dieberger et al. [11] proposed a city-based interface metaphor to highlight the challenges of spatial user interfaces, while Waller et al. [26] used urban metaphors to analyze concurrent patterns in surveillance data. Building on these prior works, we adopt the architectural metaphor to represent scientific literature, creating a “Literature City” where users can explore citation relationships in an engaging way.

3 Data Collection

To inform the design of City of Wander with a representative dataset, we focused on IEEE Transactions on Visualization and Computer Graphics (TVCG), a premier journal in the field of visualization. As a reflection of the latest advancements in visualization research, TVCG provides a rich foundation for modeling the structure and evolution of the visualization community through a metaphorical urban exploration. This study analyzed all available papers published in TVCG from 1995 to 2023. Specifically, only journal articles were included, while non-research content such as covers and editorials was excluded. Metadata, including titles, authors, abstracts, keywords, citations, publication dates, and DOIs, was collected from IEEE Xplore. Missing information was manually supplemented, and papers with incomplete or inaccessible metadata were excluded, resulting in a dataset of 3,052 papers. To enhance thematic modeling, session data from related conferences (e.g., IEEE Vis, SciVis, InfoVis, and VAST) was incorporated, as many papers from these conferences are included in TVCG. Specifically, session data for 393 TVCG papers was collected, with additional session information from the conferences used to guide the thematic structure.

4 Design of City of Wander

Building on the dataset of TVCG publications, we designed City of Wander to transform the abstract landscape of scientific literature into an engaging and intuitive exploration experience. Drawing inspiration from the metaphor of “Architecture-City”, we represent scientific documents as architectural forms within a virtual cityscape. This design integrates the themes, citation counts, and publication dates of papers into the visual and spatial attributes of buildings, offering users a tangible way to grasp complex research relationships.

To support diverse exploration needs, we designed two complementary perspectives (Figure 2): the Bird’s-Eye Perspective and

the Ground-Level Perspective (Figure 2). The *Bird’s-Eye Perspective* provides a macroscopic overview of the city, allowing users to identify patterns, distributions, and relationships. In contrast, the *Ground-Level Perspective* offers an immersive, first-person view for exploring individual buildings and accessing detailed information about specific literature.

4.1 Bird’s-Eye Perspective: Mapping Knowledge Flow and Thematic Structure

The Bird’s-Eye Perspective is designed to provide users with a high-level understanding of the knowledge system. This perspective helps readers identify thematic structures, temporal distributions, and citation relationships across a vast dataset, offering clear navigation paths during exploration. By presenting a macroscopic view of IEEE TVCG articles, it facilitates an understanding of how topics evolve and interconnect over time, ensuring that users can orient themselves within the broader landscape of visualization research.

To construct this perspective, we organized the knowledge system of IEEE TVCG articles into a radial city layout, simulating urban patterns based on themes, time, and citation relationships (Figure 3). Articles were categorized into nine topics using BERTopic [15], a framework for topic modeling that clusters articles by content similarity. Within the city, each thematic category forms a distinct sector. Articles are arranged in a spiral layout, radiating outward from the center to reflect their chronological order, symbolizing city expansion. Citation relationships between articles are represented as roads connecting buildings, forming a network of intellectual connections. To enhance spatial clarity, a force-directed algorithm adjusts the layout, clustering related articles closer together while maintaining sufficient spacing to prevent overlap. Additionally, an edge-bundling algorithm aggregates roads with similar directions to minimize visual interference from crossing lines.

4.2 Ground-Level Perspective: Immersive Exploration of Individual Knowledge

The Ground-Level Perspective focuses on deepening user engagement by fostering curiosity and enabling detailed exploration of individual documents. This perspective offers an immersive, first-person experience, where readers can navigate the city, observe architectural details up close, and access specific information about each piece of literature. The goal is to make knowledge discovery engaging and intuitive, encouraging continuous exploration of the diversity within the knowledge landscape.

In this perspective, each article is represented as a unique building, with its architectural features encoding key attributes such as theme, citation count, and publication year. Themes define the overall building forms, ensuring clear distinctions among categories for quick visual recognition by leveraging detail complexity and morphological structure. Citation count determines building height—taller structures signify higher impact—while publication year influences the building’s curvature: earlier publications are represented with simpler, linear designs, and recent ones adopt more complex, curved forms, symbolizing progression (Figure 4).

The architectural designs are generated using Grasshopper, a parametric modeling tool. First, a base model is created for each thematic category. Citation counts and publication years are then

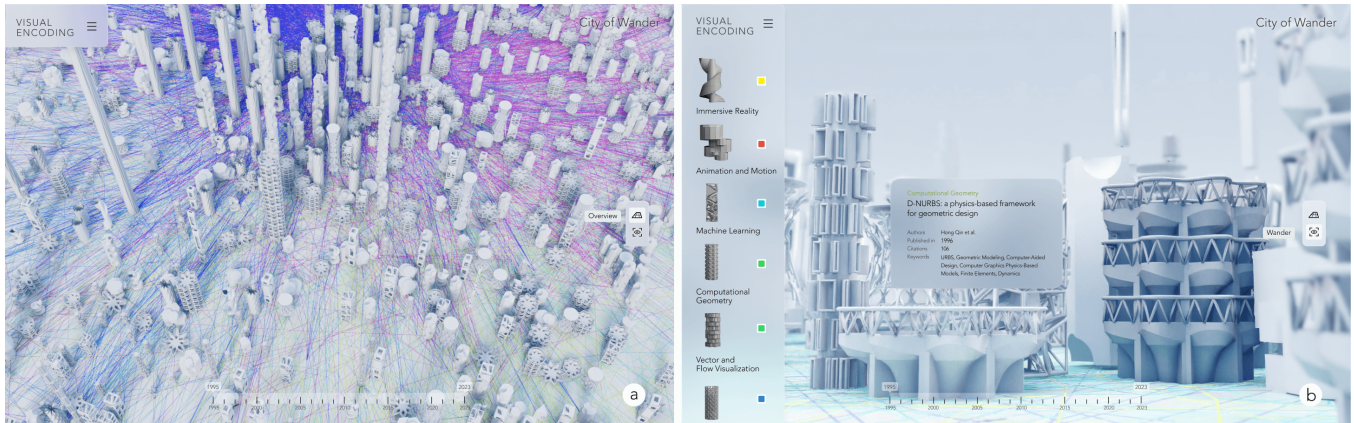


Figure 2: The interface of City of Wander, including (a) Bird's-Eye Perspective, providing a top-down perspective, and (b) Ground-Level Perspective, offering an immersive exploration experience.

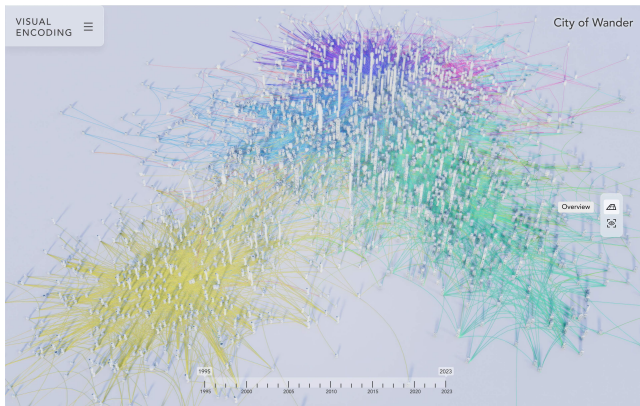


Figure 3: Distribution of buildings in the bird's eye perspective based on the citation network colored according to the cited relationship.

mapped parametrically to generate unique buildings for all articles within the city. This systematic process enables both individualized representation and consistent thematic coherence across the cityscape.

5 Case Study: Tracing the Evolution of Immersive Reality in City of Wander

This case study illustrates how City of Wander empowers users to explore the evolution of the Immersive Reality field, offering both a macroscopic view of its development and granular insights into key research trends. By leveraging the Bird's-Eye Perspective and temporal navigation tools, users can intuitively analyze the field's growth, interdisciplinary connections, and emerging research focuses.

Upon entering City of Wander, users begin in the Bird's-Eye Perspective of the city. This view provides a high-level overview of the citation network, with roads representing citation relationships

between scientific papers, clustered within a radial city layout according to themes. To focus on Immersive Reality, the user filters the city by selecting Virtual Reality from the thematic selector, isolating papers in this field and their related citation connections. Using the timeline tool, the user divides the field's development into three temporal slices—1995–2004, 2005–2014, and 2015–2023 (Figure 7)—to uncover how the field evolved over nearly three decades.

Foundations of Immersive Reality (1995–2004): The Immersive Reality field is in its infancy, with only a few scattered buildings representing publications during this time (Figure 5-a). Citation roads, colored blue and green, indicate early influences from foundational fields such as Computational Geometry and Volume Rendering. Clicking on an early paper reveals that these disciplines contributed critical methodologies to Immersive Reality's nascent development.

Expansion and Interdisciplinary Growth (2005–2014): This period marks significant expansion, with a noticeable increase in publications (Figure 5-b). The cityscape becomes denser, reflecting the field's growth and the emergence of interdisciplinary knowledge exchanges. Citation roads connect Immersive Reality to a broader range of fields, signaling the integration of diverse research approaches and technologies.

Rapid Advancements and Emerging Frontiers (2015–2023): Immersive Reality enters a phase of rapid growth, with a sharp rise in publication numbers (Figure 5-c). Buildings grow taller, symbolizing the increasing impact of individual papers, and citation networks become more cohesive, showing concentrated knowledge flows within the field. High-citation papers from this period highlight emerging trends, such as brain-computer interface technologies and comprehensive evaluations of Virtual Reality applications, which provide crucial guidance for research directions.

6 Informal Testing in the Exhibition

City of Wander was showcased at the exhibition, offering the audience an immersive VR experience (Figure 6). A brief introduction to the design concept, ideas, and illustrations was provided through videos and posters. Additionally, 3D prints of distinctive, localized

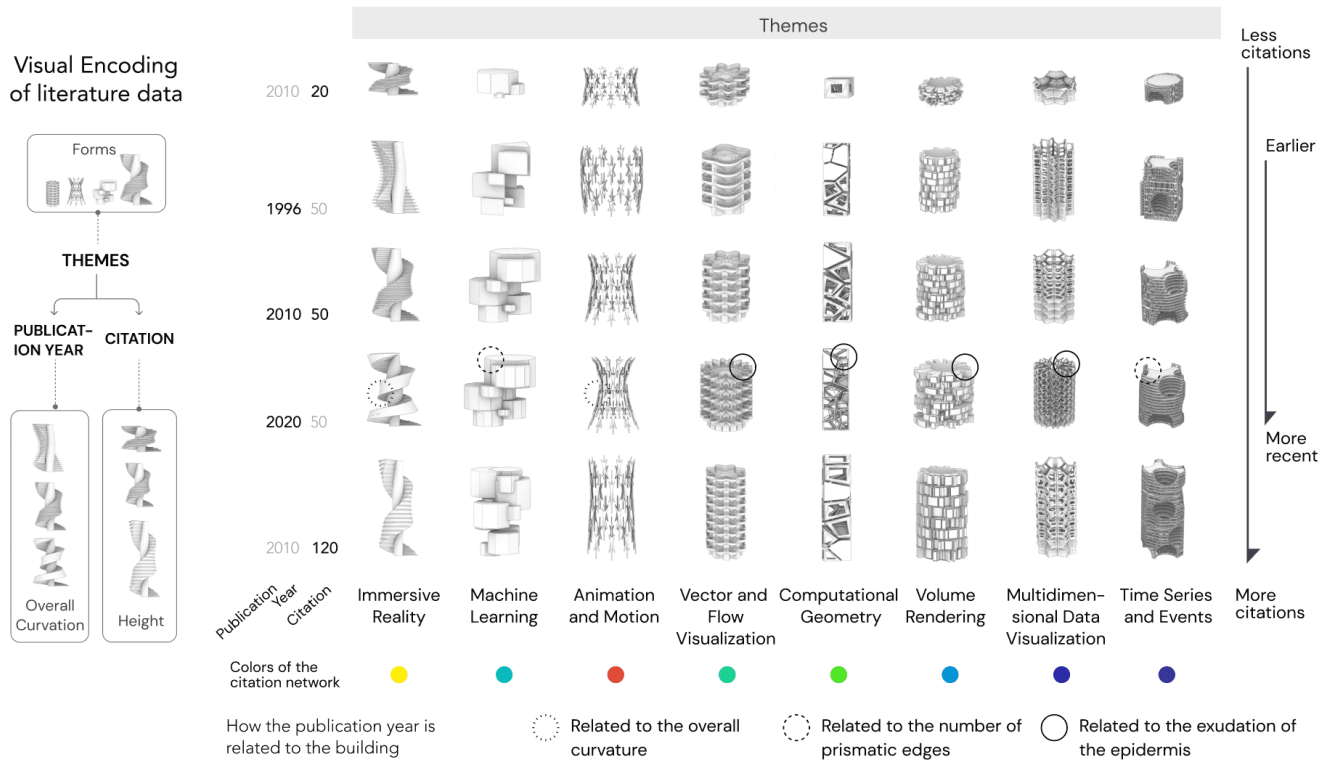


Figure 4: Visual encoding of architecture and literature data

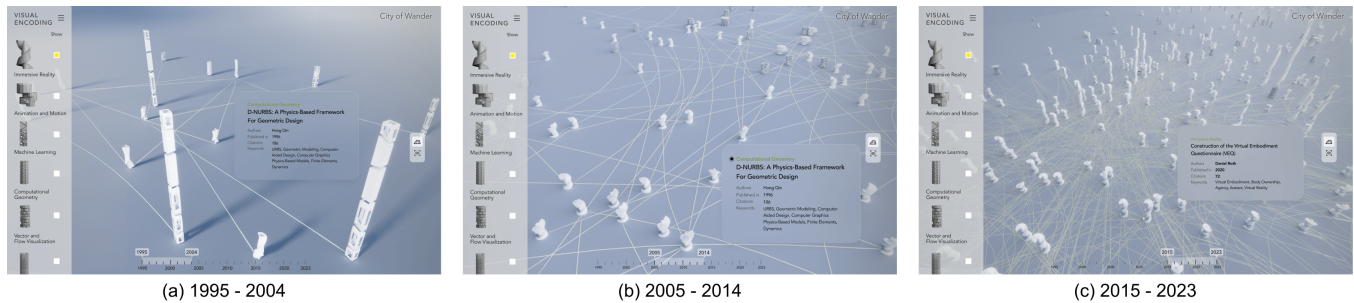


Figure 5: This visualization illustrates the evolution of the Immersive Reality field across three time periods—1995–2004, 2005–2014, and 2015–2023.

areas of the city were displayed, highlighting key features of the data. We invited non-expert audience members, who expressed interest, to explore City of Wander, aiming to observe their subjective reactions and reflections during the experience. We also gathered feedback by asking them what insights they gained from it.

Audiences expressed high interest in City of Wander, with one noting, “The correspondence between literature and architecture surprised me; I could’t believe I could read literature while walking.” Many appreciated the walking format’s positive impact on exploration and the intuitiveness of 3D models. One commented, “3D models are easier to memorize than data or text; the building height impressed me more than citation numbers.” Another mentioned: “Buildings with

varying details and shapes, like tower-like structures for immersive reality, helped me quickly distinguish research areas.” Some audiences highlighted the engaging nature of the experience, saying: “Knowledge has become a place to explore, making the process more exciting than reading a paper.” These insights illustrate the design’s potentials to spark interest, guide information discovery, and foster an open-minded approach to learning.

7 Discussion and Future Work

Visual Metaphors for Scientific Literature Visualization. The “Architecture-City” metaphor in City of Wander demonstrates how visual metaphors



Figure 6: Audiences were experiencing the City of Wander during exhibition

can make scientific literature more intriguing by transforming abstract data into familiar, intuitive forms. This approach lowers cognitive barriers, encourages engagement, and supports diverse exploration goals, especially for non-expert users. However, the current design is constrained by its fixed metaphor, which may not scale effectively across different domains or cater to the cultural and professional diversity of users. Future research should investigate adaptive metaphor frameworks that leverage AI to dynamically tailor visualizations to users' preferences and contexts. For instance, a "Garden of Knowledge" metaphor might resonate with biologists, while a "Molecular Network" metaphor could appeal to chemists. Integrating mixed-reality environments, where users can physically explore and interact with metaphorical representations, could further enhance immersion and engagement, paving the way for more inclusive and adaptable visualization systems.

The Role of Design in Knowledge Exploration. The dual-perspective design of City of Wander highlights the critical role of thoughtful design in knowledge visualization. The Bird's-Eye Perspective provides a macroscopic view of thematic structures and citation flows, while the Ground-Level Perspective offers immersive, detailed exploration of individual documents. This complementary design balances clarity with engagement, enabling users to explore scientific literature in a structured yet dynamic way. However, the lack of seamless transitions between perspectives can disrupt the user's flow, hindering their ability to connect high-level patterns with detailed insights. We propose investigating seamless perspective transitions in dual-perspective systems. For example, interactive visual cues or narrative pathways could guide users from an overview of thematic clusters to detailed information about specific articles. Additionally, incorporating aesthetic-driven design evaluations could uncover how visual styles, such as architectural motifs or urban layouts, influence user engagement and cognitive load. Further, we will conduct comprehensive user studies to compare the effectiveness with other scientific literature visualization methods in terms of learning outcomes. Combining these insights

with adaptive storytelling systems could lead to more cohesive and impactful knowledge exploration tools.

8 Conclusion

In this paper, we introduced City of Wander, an immersive system for visualizing scientific literature using the "Architecture-City" metaphor, bridging abstract citation networks and thematic relationships with intuitive urban landscapes. By offering dual perspectives—Bird's-Eye Perspective for thematic overviews and Ground-Level Perspective for detailed exploration—City of Wander empowers users, especially non-experts, to navigate and comprehend complex knowledge effectively. Informal testing during an exhibition highlighted the system's potential to make scientific literature more intriguing, memorable, and engaging, as participants praised its intuitive 3D models and immersive exploration experience. City of Wander demonstrates how design can enhance curiosity and learning, while pointing to opportunities for adaptive metaphors, perspective transitions, and mixed-reality applications in future research.

Acknowledgments

Yang Shi is the corresponding author. This work was supported by Youth Fund for Humanities and Social Sciences Research, Ministry of Education 24YJCZH168 and China Center for International People-to-People Exchange CCIPE-YXSJ-20240005.

References

- [1] Bon Adriel Aseniero, George Fitzmaurice, Sheelagh Carpendale, and Justin Matejka. 2022. Skyglyphs: Reflections on the design of a delightful visualization. In 2022 IEEE VIS Arts Program (VISAP). IEEE, 105–120.
- [2] Michael Böttinger, Helen-Nicole Kostis, Maria Velez-Rojas, Penny Rheingans, and Anders Ynnerman. 2020. Reflections on visualization for broad audiences. *Foundations of data visualization* (2020), 297–305.
- [3] Brian Castellani. 2019. Map of the complexity sciences. (2019).
- [4] Chaomei Chen. 2006. CiteSpace II: Detecting and visualizing emerging trends and transient patterns in scientific literature. *Journal of the American Society for information Science and Technology* 57, 3 (2006), 359–377.
- [5] Shuai Chen, Sihang Li, Siming Chen, and Xiaoru Yuan. 2019. R-map: A map metaphor for visualizing information reposting process in social media. *IEEE transactions on visualization and computer graphics* 26, 1 (2019), 1204–1214.

- [6] ZiYi Chen, WenXi Wang, ZiYang Li, JinYi Zhao, and ZiYue Chen. 2024. Research on Strategies of Virtual Reality Technology to Promote Astronomy Science Popularization Education in Primary Schools. In *International Conference on Human-Computer Interaction*. Springer, 129–141.
- [7] J-K Chou and C-K Yang. 2011. PaperVis: Literature review made easy. In *Computer Graphics Forum*, Vol. 30. Wiley Online Library, 721–730.
- [8] Pedro Cruz. 2015. Wrongfully right: applications of semantic figurative metaphors in information visualization. *IEEE VIS Arts Program (VISAP)* (2015), 14–21.
- [9] Pedro Cruz, John Wihbey, Avni Ghael, Felipe Shibuya, and Stephen Costa. 2019. Dendrochronology of US immigration. *Information Design Journal* 25, 1 (2019), 6–20.
- [10] Weiwei Cui, Yingcai Wu, Shixia Liu, Furu Wei, Michelle X Zhou, and Huamin Qu. 2010. Context preserving dynamic word cloud visualization. In *2010 IEEE Pacific Visualization Symposium (PacificVis)*. IEEE, 121–128.
- [11] Andreas Dieberger and Andrew U Frank. 1998. A city metaphor to support navigation in complex information spaces. *Journal of Visual Languages & Computing* 9, 6 (1998), 597–622.
- [12] Marian Dörk, Nathalie Henry Riche, Gonzalo Ramos, and Susan Dumais. 2012. Pivotpaths: Strolling through faceted information spaces. *IEEE transactions on visualization and computer graphics* 18, 12 (2012), 2709–2718.
- [13] Alper Durukan, Huseyin Artun, and Atilla Temur. 2020. Virtual reality in science education: A descriptive review. *Journal of science learning* 3, 3 (2020), 132–142.
- [14] Tommaso Elli, Adam Bradley, Christopher Collins, Uta Hinrichs, Zachary Hills, and Karen Kelsky. 2020. Tied in knots: A case study on anthropographic data visualization about sexual harassment in the academy. In *2020 IEEE VIS Arts Program (VISAP)*. IEEE, 29–44.
- [15] Maarten Grootendorst. 2022. BERTopic: Neural topic modeling with a class-based TF-IDF procedure. *arXiv preprint arXiv:2203.05794* (2022).
- [16] Florian Heimerl, Qi Han, Steffen Koch, and Thomas Ertl. 2015. CiteRivers: Visual analytics of citation patterns. *IEEE transactions on visualization and computer graphics* 22, 1 (2015), 190–199.
- [17] DAVID KUTÁK. [n. d.]. Virtual Environments in the Context of Biochemical Visualization. ([n. d.]).
- [18] Outi-Marja Latvala, Tommi Keränen, Sami Noponen, Niko Lehto, Mirko Sailio, Mikko Valtu, and Pia Olli. 2017. Visualizing network events in a muggle friendly way. In *2017 International Conference On Cyber Situational Awareness, Data Analytics And Assessment (Cyber SA)*. IEEE, 1–4.
- [19] Zhanti Liang, Yongkang Xing, Kexin Guan, Zheng Da, Jianwen Fan, and Gan Wu. 2021. Design virtual reality simulation system for epidemic (COVID-19) education to public. In *Proceedings of the 4th International Conference on Control and Computer Vision*. 147–152.
- [20] Justin Matejka, Tovi Grossman, and George Fitzmaurice. 2012. Citeology: visualizing paper genealogy. In *CHI'12 extended abstracts on human factors in computing systems*. 181–190.
- [21] Shenhao Shang. 2024. An Exploration of Design Strategies for Public-Oriented Scientific Visualization Communication. *Advances in Humanities and Modern Education Research* 1, 1 (2024), 122–128.
- [22] John Stasko, Jaegul Choo, Yi Han, Mengdie Hu, Hannah Pileggi, Ramik Sadana, and Charles D Stolper. 2013. Citevis: Exploring conference paper citation data visually. *Posters of IEEE InfoVis 2* (2013).
- [23] Sonia Stephens. 2012. Placing Birds On A Dynamic Evolutionary Map: Using Digital Tools To Update The Evolutionary Metaphor Of The "Tree Of Life". (2012).
- [24] Marie B Synnæstvedt, Chaomei Chen, and John H Holmes. 2005. CiteSpace II: visualization and knowledge discovery in bibliographic databases. In *AMIA annual symposium proceedings*, Vol. 2005. American Medical Informatics Association, 724.
- [25] Nees Jan Van Eck and Ludo Waltman. 2014. CitNetExplorer: A new software tool for analyzing and visualizing citation networks. *Journal of informetrics* 8, 4 (2014), 802–823.
- [26] Jan Waller, Christian Wulf, Florian Fittkau, Philipp Döhring, and Wilhelm Haselbring. 2013. Synchronis: 3d visualization of monitoring traces in the city metaphor for analyzing concurrency. In *2013 First IEEE Working Conference on Software Visualization (VISOFT)*. IEEE, 1–4.
- [27] Yong Wang, Conglei Shi, Liangyue Li, Hanghang Tong, and Huamin Qu. 2018. Visualizing research impact through citation data. *ACM Transactions on Interactive Intelligent Systems (TiiS)* 8, 1 (2018), 1–24.